

**Lemma.** Let  $X: \mathcal{U} \subset \mathbb{R}^2 \rightarrow S$  be a parametrization of a regular surface  $S$ .  
Given  $q \in \mathcal{U}$ ,

$$DX_q[\mathbb{R}^2] = \{ \alpha'(0) \mid \alpha: (-\varepsilon, \varepsilon) \rightarrow S \text{ is differentiable and } \alpha(0) = X(q) \}.$$

**Proof.** Let  $w$  be a tangent vector at  $X(q)$ . That is,  
for some smooth curve  $\alpha: (-\varepsilon, \varepsilon) \rightarrow X[\mathcal{U}] \subset S$  such that  $\alpha(0) = X(q)$ ,  $w = \alpha'(0)$ .  
Since  $X^{-1}: X[\mathcal{U}] \rightarrow \mathbb{R}^2$  is differentiable, the curve  $\beta = X^{-1} \circ \alpha: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}$  is also differentiable, so  $DX_q \beta'(0) = DX_q \cdot DX_{X(q)}^{-1} \cdot \alpha'(0) = \alpha'(0)$ .

Conversely, let  $w = DX_q(v)$  for some  $v \in \mathbb{R}^2$ . Then, a curve

$$\gamma: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}: t \mapsto tV + q$$

satisfies that  $\gamma'(0) = V$  and  $\gamma(0) = q$ . Now, a curve  $\alpha = X \circ \gamma$  satisfies  $\alpha(0) = X(q)$  and  $\alpha'(0) = DX_q \cdot V = w$ , so proved.  $\square$

**Def.** Let  $S \subset \mathbb{R}^3$  be a regular surface and  $p \in S$  be a point.  
Then, for a parametrization  $X: \mathcal{U} \rightarrow S$  at  $p \in S$ , a unit vector

$$N(q) = \frac{X_u \times X_v}{|X_u \times X_v|}(q) \quad \text{where } q = X^{-1}(p)$$

is called an unit normal vector at  $p$ .

Def. Given a regular surface  $S$  and a point  $p \in S$ , by definition, there is a parametrization  $X: \mathcal{U} \rightarrow S$  containing  $p$ , ( $p \in X[\mathcal{U}]$ ). Then, for  $q = X^{-1}(p) \in \mathcal{U}$ , the Jacobian  $dX_q: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is linear with rank 2. The plane  $dX_q[\mathbb{R}^2]$  is called a tangent plane at  $p$  for  $S$ , denote  $T_p S$ .

The tangent plane  $T_p S$  is independent of choice of the parametrization  $X$ . But, given a parametrization  $X$ ,  $T_p S$  is determined by a basis

$$\left\{ \frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q) \right\}, \text{ it is denoted simply } \{X_u, X_v\}.$$

Given a parametrization  $X: \mathcal{U} \rightarrow S$  at  $p \in S$ , determine a vector  $w \in T_p S$ .

Given  $w \in T_p S$ , there is a curve  $\alpha: (-\epsilon, \epsilon) \rightarrow X[\mathcal{U}]$  satisfying  $\alpha(0) = p, \alpha'(0) = w$ . Let  $\beta = X^{-1} \circ \alpha: (-\epsilon, \epsilon) \rightarrow \mathcal{U}$ . It is also smooth, and  $\alpha = X \circ \beta$ . Now,

$$\alpha'(0) = \frac{d}{dt}(X \circ \beta)(0) = \frac{d}{dt}X(u(t), v(t))(0) = X_u(q) \cdot u'(0) + X_v(q) \cdot v'(0)$$

where  $q = X^{-1}(p)$  and  $\beta(t) = (u(t), v(t))$ .

Let  $S$  be a regular surface and  $V \subseteq \mathbb{R}^3$  be an open set such that  $S \subseteq V$ .  
If  $f: V \rightarrow \mathbb{R}$  is differentiable, then for any point  $p \in S$  and  
parametrization  $X: U \rightarrow S$  at  $p \in S$ , a map  $f \circ X: U \rightarrow \mathbb{R}$  is differentiable.

Height function.

Given a surface  $S$  and a fixed unit vector  $V \in \mathbb{R}^3$ , a function

$h: S \rightarrow \mathbb{R}; p \mapsto \langle p, V \rangle$  is called a height function with respect to  $V \in \mathbb{R}^3$ .

Unit Sphere.

Let  $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$  be given. Define a map

$$R_{z, \theta}: \mathbb{R}^3 \rightarrow \mathbb{R}^3; (x, y, z) \mapsto (\cos \theta \cdot x - \sin \theta \cdot y, \sin \theta \cdot x + \cos \theta \cdot y, z).$$

(That is,  $Z$ -rotation of degree  $\theta$ ). Since  $\cos \theta, \sin \theta$  are just constant,  $R_{z, \theta}$  is differentiable. So, the restriction  $R_{z, \theta}|_{S^2}$  is also differentiable.

Calculate the tangent plane at a point  $p \in S^2$ . Simply, denote  $R = R_{z, \theta}$ . Since  $S^2$  is regular surface, there is a smooth curve  $\alpha: (-\epsilon, \epsilon) \rightarrow S^2$  such that  $\alpha(0) = p$ .

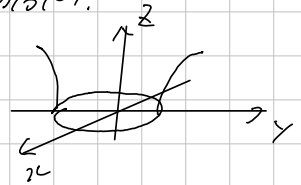
Problems, do Comma sec 2.4.

Explicit the tangent plane of  $x^2 + y^2 - z^2 = 1$  at  $(x_0, y_0, 0)$ .

Suppose that  $x_0, y_0 > 0$ . Consider a patch

$$X: D \rightarrow S: (u, v) \mapsto (\sqrt{1-u^2+v^2}, u, v)$$

where  $D$  is unit disk.



Exercise,

Sec 2.4, 11.

Show that a normal line of a surface parametrized by

$$X(u, v) = (f(u) \cdot \cos v, f(u) \cdot \sin v, g(u)), \quad f(u) \neq 0, \quad g'(u) \neq 0.$$

must cross the Z-axis.

Proof. Since the normal vector is determined by: for each  $q = (u, v)$ ,

$$N(q) = (X_u \times X_v)(q) \quad \text{and} \quad X_u = (f'(u) \cos v, f'(u) \sin v, g'(u))$$

$$X_v = (-f(u) \sin v, f(u) \cos v, 0).$$

So

$$N(q) = (g'(u)f(u)\cos v, -g'(u)f(u)\sin v, f'(u)f(u)).$$

Now, the normal line is

$$L(t) = X(q) + t \cdot N(q) = (f(u)\cos v(1-tg'(u)), f(u)\sin v(1-tg'(u)), g(u) + tf'(u)f(u))$$

cross the Z-Axis when  $t = 1/g'(u)$ .